Document Title

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1 Introduction

1.1 Matérn covariance function and its parametrizations

From [2], the Matérn covariance function is defined as

$$r(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\sqrt{2\nu}h}{\rho} \right)^{\nu} K_{\nu} \left(\frac{\sqrt{2\nu}h}{\rho} \right) \tag{1}$$

and its corresponding spectral density is

$$\hat{f}(\omega) = \frac{\sigma^2 \sqrt{4\pi}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + 4\pi^2\omega^2 \right)^{-(\nu+1/2)} \tag{2}$$

computed by using the Fourier transform pair parametrization

$$\hat{f}(\omega) = \int_{\mathbb{R}} e^{-i2\pi\omega h} r(h) dh \iff r(h) = \int_{\mathbb{R}} e^{i2\pi\omega h} \hat{f}(\omega) d\omega \tag{3}$$

But actually we want the pair parametrization

$$\tilde{f}(\omega) = \int_{\mathbb{R}} e^{-i\omega h} r(h) dh \iff r(h) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{i\omega h} \tilde{f}(\omega) d\omega \tag{4}$$

For that, we simply do

$$\tilde{f}(\omega) = \hat{f}\left(\frac{\omega}{2\pi} \right) = \frac{\sigma^2 \sqrt{4\pi}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{5}$$

But actually we apparently want this Fourier pair parametrization

$$\overline{f}(\omega) = \mathfrak{F}[r(h)](\omega) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{-i\omega h} r(h) dh \iff r(h) = \mathfrak{F}^{-1}[\overline{f}(\omega)](h) = \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{i\omega h} \overline{f}(\omega) d\omega \tag{6}$$

For that, we simply do

$$\overline{f}(\omega) = \frac{1}{\sqrt{2\pi}} \tilde{f}(\omega) = \frac{\sigma^2 \sqrt{2}\Gamma(\nu + 1/2)}{\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{7}$$

Recall the scaling property of the Fourier transform given by

$$r(ah) \iff \frac{1}{|a|} \overline{f}\left(\frac{\omega}{a} \right) \tag{8}$$

Now let $\varrho(h) = r(2h)$ and $\kappa = \sqrt{8\nu}/\rho$, then

$$\varrho(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} \left(\frac{\sqrt{8\nu}h}{\rho} \right)^{\nu} K_\nu \left(\frac{\sqrt{8\nu}h}{\rho} \right) \tag{9}$$

$$= \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{10}$$

which is the familiar form of the Matérn covariance function. Just for the record, if we have it in terms of τ, then

$$\varrho(h) = \frac{1}{2^{\nu-1}\Gamma(\nu)} \frac{\Gamma(\nu)}{\Gamma(\nu + 1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{11}$$

$$= \frac{1}{2^{\nu-1}\Gamma(\nu + 1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2} (\kappa h)^{\nu} K_\nu(\kappa h) \tag{12}$$

Setting $\nu = \frac{1}{2}$, we obtain

$$\varrho(h) = \frac{1}{2^{-1/2}\Gamma(1)\sqrt{4\pi}\kappa^{1}\tau^2} (\kappa h)^{1/2} K_{1/2}(\kappa h). \tag{13}$$

Using the identities

$$\Gamma(1) = 1, \tag{14}$$

$$K_{1/2}(x) = \sqrt{\frac{\pi}{2x}} e^{-x}, \tag{15}$$

we get

$$(\kappa h)^{1/2} K_{1/2}(\kappa h) = (\kappa h)^{1/2} \sqrt{\frac{\pi}{2\kappa h}} e^{-\kappa h} = \sqrt{\frac{\pi}{2}}\, e^{-\kappa h}. \tag{16}$$

Substituting back,

$$\varrho(h) = \frac{1}{2^{-1/2}\sqrt{4\pi}\kappa\tau^2} \sqrt{\frac{\pi}{2}}\, e^{-\kappa h}. \tag{17}$$

Now simplify the constants:

$$2^{-1/2}\sqrt{4\pi} = \frac{1}{\sqrt{2}} \cdot 2\sqrt{\pi} = \sqrt{2\pi}, \tag{18}$$

so that

$$\varrho(h) = \frac{1}{\sqrt{2\pi}\,\kappa\tau^2} \sqrt{\frac{\pi}{2}}\, e^{-\kappa h} = \frac{1}{2\kappa\tau^2} e^{-\kappa h}. \tag{19}$$

By the scaling property, the corresponding spectral density to $\varrho(h)$ is $f(\omega) = \frac{1}{2}\overline{f}\left(\frac{\omega}{2}\right)$ given by

$$f(\omega) = \frac{\sigma^2\sqrt{2}\Gamma(\nu + 1/2)}{2\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{2\nu}{\rho^2} + \frac{\omega^2}{4} \right)^{-(\nu+1/2)} \tag{20}$$

$$= \frac{\sigma^2\sqrt{2}\Gamma(\nu + 1/2)}{2\Gamma(\nu)} \left(\frac{2\nu}{\rho^2} \right)^{\nu} \left(\frac{1}{4} \right)^{-(\nu+1/2)} \left(\frac{8\nu}{\rho^2} + \omega^2 \right)^{-(\nu+1/2)} \tag{21}$$

Since $\left(\frac{1}{4}\right)^{-(\nu+1/2)} = 2 \cdot 4^{\nu}$ and $\kappa^2 = 8\nu/\rho^2$ we have that

$$f(\omega) = \frac{\sigma^2\sqrt{2}\Gamma(\nu+1/2)}{\Gamma(\nu)}\left(\frac{8\nu}{\rho^2}\right)^{\nu}\left(\frac{8\nu}{\rho^2}+\omega^2\right)^{-(\nu+1/2)} \tag{22}$$

$$= \frac{\sigma^2\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \tag{23}$$

Let $A = A_{(\nu,\kappa)} = \dfrac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}$ so this matches with [1]. If $\sigma^2 = \dfrac{\Gamma(\nu)}{\Gamma(\nu+1/2)\sqrt{4\pi}\kappa^{2\nu}\tau^2}$, then $A\sigma^2 = (2\pi)^{-1/2}\tau^{-2}$. The moral of the story is that

$$f(\omega) = A\sigma^2\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \iff \varrho(h) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^{\nu}K_{\nu}(\kappa h), \tag{24}$$

Putting in terms of α, we instead have

$$f(\omega) = A\sigma^2\left(\kappa^2+\omega^2\right)^{-\alpha} \iff \varrho(h) = \frac{\sigma^2}{2^{\alpha-1/2-1}\Gamma(\alpha-1/2)}(\kappa h)^{\alpha-1/2}K_{\alpha-1/2}(\kappa h), \tag{25}$$

If we drop σ^2, we still have the Fourier pair

$$f_\star(\omega) = A\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \iff \varrho_\star(h) = \frac{1}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^{\nu}K_{\nu}(\kappa h), \tag{26}$$

1.2 Computational quirks

The `MetricGraph` package actually codifies

$$\overline{\varrho}(h;\kappa,\nu,\texttt{cte}) = \frac{\texttt{cte}^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|) \tag{27}$$

So if you want something like

$$\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\varrho(h;\kappa,\nu,\sigma) = \frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|) \tag{28}$$

computationally you should actually do

$$\sigma^2\overline{\varrho}\left(h;\kappa,\nu,\sqrt{\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}}\right) \tag{29}$$

1.3 Preliminaries

We consider the Matérn covariance function

$$\varrho(h;\kappa,\nu,\sigma) = \frac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa|h|)^\nu K_\nu(\kappa|h|), \quad \sigma^2 = A^{-1}(2\pi)^{-1/2}\tau^{-2}, \tag{30}$$

where $\rho = \sqrt{8\nu}/\kappa$ and $\nu = \alpha - 1/2$ and the SPDE

$$(\kappa^2 - \Delta_\Gamma)^{\alpha/2}(\tau u) = \mathcal{W}, \qquad \text{on } \Gamma, \tag{31}$$

1.4 Computing spectral density from SPDE

We take the Fourier transform (6) to (31) to get

$$(\kappa^2+\omega^2)^{\alpha/2}\tau\hat{u}(\omega) = \widehat{\mathcal{W}}(\omega) \implies f(\omega) = \frac{1}{\tau^2\sqrt{2\pi}}(\kappa^2+\omega^2)^{-\alpha} \tag{32}$$

Let $L = (\kappa^2 - \Delta_\Gamma)^{\alpha/2}$. The solution u is a centered Gaussian random field with covariance operator $\mathcal{C} = \tau^{-2}L^{-\alpha}$. We consider the approximation

$$\mathcal{C}_\alpha = \tau^{-2}L^{-\lfloor\alpha\rfloor}P_m(L^{-1})Q_m(L^{-1})^{-1} \tag{33}$$

where

$$x^{\{\alpha\}} \approx \frac{P_m(x)}{Q_m(x)} = \frac{\sum_{i=0}^m a_i x^i}{\sum_{j=0}^m b_j x^j} = \frac{p_m(x^{-1})}{q_m(x^{-1})} \tag{34}$$

or

$$x^{-\{\alpha\}} \approx \frac{P_m(x^{-1})}{Q_m(x^{-1})} = \frac{\sum_{i=0}^m a_i x^{m-i}}{\sum_{j=0}^m b_j x^{m-j}} = \frac{p_m(x)}{q_m(x)} \tag{35}$$

We further have that

$$\frac{p_m(x)}{q_m(x)} = \frac{P_m(x^{-1})}{Q_m(x^{-1})} = k + \sum_{i=1}^m r_i(x-p_i)^{-1}. \tag{36}$$

Therefore

$$\mathcal{C}_\alpha = \tau^{-2}L^{-\lfloor\alpha\rfloor}\left(kI + \sum_{i=1}^m r_i(L-p_iI)^{-1}\right) = \tau^{-2}kL^{-\lfloor\alpha\rfloor} + \sum_{i=1}^m \tau^{-2}r_iL^{-\lfloor\alpha\rfloor}(L-p_iI)^{-1} \tag{37}$$

Since $k, r_i > 0$ and $p_i < 0$, $\mathcal{C}_0 = \tau^{-2} k L^{-\lfloor \alpha \rfloor}$ and $\mathcal{C}_i = \tau^{-2} r_i L^{-\lfloor \alpha \rfloor}(L - p_i I)^{-1}$ are valid covariance operators so that $\mathcal{C}_\alpha = \mathcal{C}_0 + \sum_{i=1}^m \mathcal{C}_i$. Then the approximation u_α has decomposition

$$u_\alpha = u_0 + \sum_{i=1}^m u_i \tag{38}$$

where $u_0 \sim N(0, \mathcal{C}_0)$ and $u_i \sim N(0, \mathcal{C}_i)$. Moreover, they satisfy

$$\mathcal{C}_0^{-1/2} u_0 = L^{\lfloor \alpha \rfloor / 2}(\tau k^{-1/2} u_0) = \mathcal{W}, \qquad \text{on } \Gamma, \tag{39}$$

and

$$\mathcal{C}_i^{-1/2} u_i = (L^{\lfloor \alpha \rfloor}(L - p_i I))^{1/2}(\tau r_i^{-1/2} u_i) = \mathcal{W}, \qquad \text{on } \Gamma \tag{40}$$

Definition 1.1. u_i in (40) is called a shifted Whittle-Matérn process on Γ.

Remark 1. Observe that (39) are Whittle-Matérn process where the power $\lfloor \alpha \rfloor$ is integer.

Observe now that (30) has spectral density

$$f(\omega) = A\sigma^2(\kappa^2 + \omega^2)^{-(\nu + 1/2)}, \quad A = \sigma^{-2}(2\pi)^{-1/2}\tau^{-2} \tag{41}$$

which can be also written as

$$f(\omega) = A\sigma^2\kappa^{-2\alpha}(1 + \kappa^{-2}\omega^2)^{-\alpha} = A\sigma^2\kappa^{-2\alpha}(1 + \kappa^{-2}\omega^2)^{-\lfloor \alpha \rfloor}(1 + \kappa^{-2}\omega^2)^{-\{\alpha\}} \tag{42}$$

By (35), we can approximate f by f_α given by

$$f_\alpha(\omega) = A\sigma^2\kappa^{-2\alpha}\frac{1}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}}\frac{p_m(1 + \kappa^{-2}\omega^2)}{q_m(1 + \kappa^{-2}\omega^2)} \tag{43}$$

By (36), we have that

$$f_\alpha(\omega) = A\sigma^2\kappa^{-2\alpha}\frac{1}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}}\left(k + \sum_{i=1}^m r_i(1 + \kappa^{-2}\omega^2 - p_i)^{-1}\right) \tag{44}$$

$$= \frac{A\sigma^2\kappa^{-2\alpha}k}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}} + \sum_{i=1}^m \frac{A\sigma^2\kappa^{-2\alpha}r_i}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}(1 + \kappa^{-2}\omega^2 - p_i)} \tag{45}$$

Let

$$f_0(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}k}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}} \quad \text{and} \quad f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}r_i}{(1 + \kappa^{-2}\omega^2)^{\lfloor \alpha \rfloor}(1 + \kappa^{-2}\omega^2 - p_i)} \tag{46}$$

then we can write

$$f_\alpha(\omega) = f_0(\omega) + \sum_{i=1}^m f_i(\omega), \tag{47}$$

which is a valid decomposition of spectral densities. We now can equivalently write

$$f_0(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}k\kappa^{2\lfloor \alpha \rfloor}}{(\kappa^2 + \omega^2)^{\lfloor \alpha \rfloor}} \quad \text{and} \quad f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha}r_i\kappa^{2\lfloor \alpha \rfloor + 2}}{(\kappa^2 + \omega^2)^{\lfloor \alpha \rfloor}(\kappa^2(1 - p_i) + \omega^2)} \tag{48}$$

Now for $k \in \mathbb{N}$ and $p \in \mathbb{R}$, we have that

$$\frac{1}{(\kappa^2 + \omega^2)^k(\kappa^2(1 - p) + \omega^2)} = \frac{1}{(\kappa^2 p)^k(\kappa^2(1 - p) + \omega^2)} - \sum_{j=1}^k \frac{1}{(\kappa^2 p)^{k+1-j}(\kappa^2 + \omega^2)^j} \tag{49}$$

Indeed, let $r = \dfrac{\kappa^2 p}{\kappa^2+\omega^2}$, then $1-r = \dfrac{\kappa^2(1-p)+\omega^2}{\kappa^2+\omega^2}$. Hence

$$\begin{aligned}
\sum_{j=1}^{k} \frac{1}{(\kappa^2 p)^{k+1-j}(\kappa^2+\omega^2)^j} &= \frac{1}{(\kappa^2 p)^{k+1}} \sum_{j=1}^{k} \left(\frac{\kappa^2 p}{\kappa^2+\omega^2}\right)^j \\
&= \frac{1}{(\kappa^2 p)^{k+1}} \frac{r(1-r^k)}{1-r} \\
&= \frac{1}{(\kappa^2 p)^{k+1}} \frac{\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\left(1-\left(\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\right)^k\right)}{\dfrac{\kappa^2(1-p)+\omega^2}{\kappa^2+\omega^2}} \\
&= \frac{1-\left(\dfrac{\kappa^2 p}{\kappa^2+\omega^2}\right)^k}{(\kappa^2 p)^k(\kappa^2(1-p)+\omega^2)} \\
&= \frac{1}{(\kappa^2 p)^k(\kappa^2(1-p)+\omega^2)} - \frac{1}{(\kappa^2+\omega^2)^k(\kappa^2(1-p)+\omega^2)}
\end{aligned}$$

Therefore

$$f_i(\omega) = \frac{A\sigma^2\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor}(\kappa^2(1-p_i)+\omega^2)} \tag{50}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A\sigma^2\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j}(\kappa^2+\omega^2)^j} \tag{51}$$

$$= \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor}} \sigma^2(\kappa^2(1-p_i)+\omega^2)^{-1} \tag{52}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j}} \sigma^2(\kappa^2+\omega^2)^{-j} \tag{53}$$

$$= \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor} A_{(1/2,\kappa\sqrt{1-p_i})}} A_{(1/2,\kappa\sqrt{1-p_i})} \sigma^2(\kappa^2(1-p_i)+\omega^2)^{-(1/2+1/2)} \tag{54}$$

$$- \sum_{j=1}^{\lfloor\alpha\rfloor} \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha} r_i \kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2 p_i)^{\lfloor\alpha\rfloor+1-j} A_{(j-1/2,\kappa)}} A_{(j-1/2,\kappa)} \sigma^2(\kappa^2+\omega^2)^{-(j-1/2+1/2)} \tag{55}$$

$$= f_{i0}(\omega) - \sum_{j=1}^{\lfloor\alpha\rfloor} f_{ij}(\omega) \tag{56}$$

$$\begin{cases}
\varrho(h) & = \mathfrak{F}^{-1}[f(\omega)](h) = \dfrac{\sigma^2}{2^{\nu-1}\Gamma(\nu)}(\kappa h)^\nu K_\nu(\kappa h) \\
f(\omega) & = A_{(\nu,\kappa)}\sigma^2\left(\kappa^2+\omega^2\right)^{-(\nu+1/2)} \\
A_{(\nu,\kappa)} & = \dfrac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)} = \dfrac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)} \\
\sigma^2 & = A_{(\nu,\kappa)}^{-1}(2\pi)^{-1/2}\tau^{-2} \\
\rho & = \sqrt{8\nu}/\kappa \\
\nu & = \alpha - 1/2
\end{cases} \tag{57}$$

Similarly for f_0

$$f_0(\omega) = A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}\sigma^2(\kappa^2+\omega^2)^{-\lfloor\alpha\rfloor} = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}A_{(\lfloor\alpha\rfloor-1/2,\kappa)}\sigma^2(\kappa^2+\omega^2)^{-(\lfloor\alpha\rfloor-1/2+1/2)} \tag{58}$$

By taking the inverse Fourier transform, we arrive at

$$\varrho_0(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}k\kappa^{2\lfloor\alpha\rfloor}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,\sigma) \tag{59}$$

and

$$\varrho_i(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor}A_{(1/2,\kappa\sqrt{1-p_i})}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) - \sum_{j=1}^{\lfloor\alpha\rfloor}\frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor+1-j}A_{(j-1/2,\kappa)}}\varrho(h;\kappa,j-1/2,\sigma) \tag{60}$$

Let us simplify the expressions above. We first simplify ϱ_0.

$$\begin{aligned}
\varrho_0(h) &= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\frac{A_{(\nu,\kappa)}}{A_{(\lfloor\alpha\rfloor-1/2,\kappa)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\frac{\frac{\sqrt{2}\Gamma(\nu+1/2)\kappa^{2\nu}}{\Gamma(\nu)}}{\frac{\sqrt{2}\Gamma(\lfloor\alpha\rfloor-1/2+1/2)\kappa^{2(\lfloor\alpha\rfloor-1/2)}}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor}\kappa^{2\nu}\kappa^{-2(\lfloor\alpha\rfloor-1/2)}\frac{\frac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\frac{\Gamma(\lfloor\alpha\rfloor)}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+\nu-\lfloor\alpha\rfloor+1/2)}\frac{\frac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\frac{\Gamma(\lfloor\alpha\rfloor)}{\Gamma(\lfloor\alpha\rfloor-1/2)}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\frac{c_\alpha}{c_{\lfloor\alpha\rfloor}}\varrho(h;\kappa,\lfloor\alpha\rfloor-1/2,1)\\
&= k\sigma^2\varrho\left(h;\kappa,\lfloor\alpha\rfloor-1/2,\sqrt{c_\alpha/c_{\lfloor\alpha\rfloor}}\right)
\end{aligned}$$

where $c_a = \Gamma(a)/\Gamma(a-1/2)$. For simplicity, let

$$\varrho_{i0}(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor}A_{(1/2,\kappa\sqrt{1-p_i})}}\varrho(h;\kappa\sqrt{1-p_i},1/2,\sigma) \tag{61}$$

and

$$\varrho_{ij}(h) = \frac{A_{(\nu,\kappa)}\kappa^{-2\alpha}r_i\kappa^{2\lfloor\alpha\rfloor+2}}{(\kappa^2p_i)^{\lfloor\alpha\rfloor+1-j}A_{(j-1/2,\kappa)}}\varrho(h;\kappa,j-1/2,\sigma) \tag{62}$$

Let us now simplify ϱ_{i0}.

$$\begin{aligned}
\varrho_{i0}(h) &= r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor}}\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor+2}\kappa^{-2\lfloor\alpha\rfloor}\frac{A_{(\nu,\kappa)}}{A_{(1/2,\kappa\sqrt{1-p_i})}}\varrho(h;\kappa\sqrt{1-p_i},1/2,1)\\
&= r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor}}\kappa^{-2\alpha}\kappa^{2\lfloor\alpha\rfloor+2}\kappa^{-2\lfloor\alpha\rfloor}\frac{\frac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)}}{\frac{\sqrt{2}\Gamma(1/2+1/2)(\kappa\sqrt{1-p_i})^{2\cdot 1/2}}{\Gamma(1/2)}}\varrho(h;\kappa\sqrt{1-p_i},1/2,1)\\
&= r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor}}\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor+\nu-1/2)}\frac{\frac{\Gamma(\alpha)}{\Gamma(\alpha-1/2)}}{\frac{\Gamma(1/2+1/2)\sqrt{1-p_i}}{\Gamma(1/2)}}\varrho(h;\kappa\sqrt{1-p_i},1/2,1)\\
&= r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor}}\frac{c_\alpha\sqrt{\pi}}{\sqrt{1-p_i}}\varrho(h;\kappa\sqrt{1-p_i},1/2,1)\\
&= r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor}}\varrho\left(h;\kappa\sqrt{1-p_i},1/2,\sqrt{c_\alpha\sqrt{\pi}/\sqrt{1-p_i}}\right)
\end{aligned}$$

since $\Gamma(1)=1$ and $\Gamma(1/2)=\sqrt{\pi}$. Now we simplify ϱ_{ij}.

$$\begin{aligned}
\varrho_{ij}(h) &= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j)}\frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}}\frac{A_{(\nu,\kappa)}}{A_{(j-1/2,\kappa)}}\varrho(h;\kappa,j-1/2,1)\\
&= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j)}\frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}}\frac{\frac{\sqrt{2}\Gamma(\alpha)\kappa^{2\nu}}{\Gamma(\alpha-1/2)}}{\frac{\sqrt{2}\Gamma(j)\kappa^{2(j-1/2)}}{\Gamma(j-1/2)}}\varrho(h;\kappa,j-1/2,1)\\
&= r_i\sigma^2\kappa^{2(-\alpha+\lfloor\alpha\rfloor+1-\lfloor\alpha\rfloor-1+j+\nu-j+1/2)}\frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}}\frac{c_\alpha}{c_j}\varrho(h;\kappa,j-1/2,1)\\
&= r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}}\varrho\left(h;\kappa,j-1/2,\sqrt{c_\alpha/c_j}\right)
\end{aligned}$$

To sum up, we have arrived at

$$\varrho_\alpha(h)=\varrho_0(h)+\sum_{i=1}^{m}\varrho_i(h) \tag{63}$$

where

$$\varrho_0(h)=k\sigma^2\varrho\left(h;\kappa,\lfloor\alpha\rfloor-1/2,\sqrt{c_\alpha/c_{\lfloor\alpha\rfloor}}\right) \tag{64}$$

and

$$\varrho_i(h)=r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor}}\varrho\left(h;\kappa\sqrt{1-p_i},1/2,\sqrt{c_\alpha\sqrt{\pi}/\sqrt{1-p_i}}\right)-\sum_{j=1}^{\lfloor\alpha\rfloor}r_i\sigma^2\frac{1}{p_i^{\lfloor\alpha\rfloor+1-j}}\varrho\left(h;\kappa,j-1/2,\sqrt{c_\alpha/c_j}\right) \tag{65}$$

2 Comparison to rat int paper

3 Implementation

3.1 Case $\alpha \in (0, 1)$

$$\varrho_0(h) = k\sigma^2 \frac{c_\alpha \sqrt{4\pi}}{\kappa} 1_{[h=0]} \implies \mathbf{Q}_0 = \mathbf{I} \cdot \frac{1}{(k\sigma^2 c_\alpha \sqrt{4\pi})/\kappa} \tag{66}$$

$$\varrho_i(h) = r_i \frac{c_\alpha \sqrt{\pi}}{\sqrt{1-p_i}} \varrho(h; \kappa\sqrt{1-p_i}, 1/2, \sigma) \implies \mathbf{Q}_i = \frac{1}{(r_i c_\alpha \sqrt{\pi})/\sqrt{1-p_i}} \begin{cases} \overline{\nu} = 1/2 \\ \overline{\kappa} = \kappa\sqrt{1-p_i} \\ \overline{\tau} = \sqrt{\dfrac{\Gamma(\overline{\nu})}{\Gamma(\overline{\nu}+1/2)\sqrt{4\pi}\overline{\kappa}^{2\overline{\nu}}\sigma^2}} \\ \cdot \texttt{Qalpha1}(\overline{\tau}, \overline{\kappa}) \end{cases} \tag{67}$$

3.2 Case $\alpha \in (1, 2)$

$$\varrho_0(h) = k\frac{c_\alpha}{c_{\lfloor\alpha\rfloor}} \varrho(h; \kappa, \lfloor\alpha\rfloor - 1/2, \sigma) \implies \mathbf{Q}_0 = \frac{1}{kc_\alpha/c_{\lfloor\alpha\rfloor}} \begin{cases} \overline{\nu} = 1/2 \\ \overline{\kappa} = \kappa \\ \overline{\tau} = \sqrt{\dfrac{\Gamma(\overline{\nu})}{\Gamma(\overline{\nu}+1/2)\sqrt{4\pi}\overline{\kappa}^{2\overline{\nu}}\sigma^2}} \\ \cdot \texttt{Qalpha1}(\overline{\tau}, \overline{\kappa}) \end{cases} \tag{68}$$

$$\varrho_i(h) = \varrho_i(h) \implies \mathbf{Q}_i = \frac{1}{r_i\sigma^2} \left(\texttt{matern.p.precision}(\kappa, p_i, \alpha, \ell_e) - f(\texttt{matern.p.joint}(\kappa, p_i, \alpha))\right) \tag{69}$$

References

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